

Weather, Monitors, or Policy? Results summary

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Question. Once weather and changes in the monitoring network are removed, did PM2.5 change differently in cities enrolled in India's National Clean Air Programme (NCAP) than in comparable cities? I registered the analysis plan before making any post-2019 comparison (<https://osf.io/jksne/>). Full report, code and decision log: <https://github.com/breenu/ncap-evaluation>.

RQ1: measurement (ground monitors, descriptive).

- In the 18 NCAP cities with a PM2.5 station valid every year from 2018 to 2025 (16 with a single such station), the reported change of -23.6% becomes -14.1% after removing modelled weather and the change in which monitors exist. **H4 supported:** $+9.5$ pp (95% CI $+4.8$ to $+14.5$), of which network composition is $+5.8$ pp. For PM10 (13 cities): $+6.0$ pp ($+1.8$ to $+10.0$), mostly modelled weather. H4 holds in 31 of 31 PM2.5 versions and 29 of 31 PM10 versions.
- New stations read cleaner than existing ones in the same city and year: -6.3% raw, -6.8% deweathered (PM2.5). Across 23 cities with a 2018 panel, network composition makes the average city look 3.7 pp better by 2025 (-7.2 to -0.4).
- The audit found instrument ceilings, flatlines and 653 level shifts; missing days are *less* common on polluted days, so naive annual means are biased slightly upward ($+0.7\%$).

RQ2: weather. Weather moves a city's annual PM2.5 by a median 5.1% from one year to the next (12.6% at the 90th percentile), against a median raw year-on-year change of 10.3%.

RQ3: the satellite comparison (confirmatory). 113 urban centres containing 121 NCAP cities against 923 comparable non-NCAP centres, 2010–2024, synthetic difference-in-differences.

- Satellite PM2.5 fell in both groups after 2018: -16.8% in NCAP units, -24.3% in the comparison pool.
- **H1: not identified by this design (failed: (b) pre-trend Wald test).** NCAP units' satellite PM2.5 did not fall relative to comparable units; the estimates point to a relative rise of about 3–5%. H1 is not identified by this design, so this is not an effect of NCAP. Primary estimate $+3.6\%$ (95% CI $+2.4\%$ to $+4.8\%$); pre-trend Wald $p = 0.0008$; 17 of 19 registered checks agree, every one positive. H2 was not tested.
- Calibration leakage: the registered warning fires (gained a monitor $+2.5\%$, did not $+5.5\%$). Raw MAIAC AOD, which no monitor calibrates: $+0.8\%$ (-0.4% to $+2.0\%$) against $+4.2\%$ for satellite PM2.5 on the same units ("rise not reproduced in AOD"); the monitor-gain gap is absent ("gap absent from AOD (consistent with calibration leakage)"), though the two signals diverge most where no monitor was added (exploratory). AOD is read for direction only.
- Ground layer (secondary, one pre-year): PM2.5 before–after -14.0% , against control cities -4.7% (-13.4% to $+4.7\%$). Ground and satellite agree that these cities got cleaner; the satellite says not more than comparable cities.

RQ4: heterogeneity and mechanism (city-level relative changes, not effects of NCAP). City estimates are too noisy to rank: a typical 95% rank interval spans 75 of 113 places, and 0 pass Benjamini–Hochberg. **H5** (smaller changes in the Indo-Gangetic Plain): registered rule not met (-0.3% , -3.3% to $+2.6\%$). **H3** (dust control): inconclusive. Funding dose (exploratory): -0.4% per doubling of allocation (-3.5% to $+2.7\%$).

Key caveats. H1 is not identified by this design, so no number here is an effect of NCAP. NCAP cities are larger and less often in the Indo-Gangetic Plain than the comparison pool. Satellite PM2.5 is calibrated to ground monitors. Ground results rest mostly on single monitors and one pre-NCAP year. The confirmatory layer measures PM2.5, while NCAP's targets are for PM10. Every deviation from the registered plan is listed in Appendix A of the report.